Math 520, F15 **Name:** Answer
Quiz 4 **SSN:** xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given the initial value problem

$$y' = 1 - t + 2y, \quad y(0) = 1.$$

Use Euler 's method to find its approximate values at $t = 0.1, 0.2, 0.3$ with $h = 0.1$.

Solution: $\underline{f(t,y) = 1 - t + 2y}$, $\underline{t_0 = 0}$, $\underline{y_0 = 1}$, $\underline{h = 0.1}$.

Euler's method: $y_{n+1} = y_n + h f(t_n, y_n)$, $n = 0, 1, 2, \cdots$

$t_1 = 0.1$

$y_1 = y_0 + h f(t_0, y_0) = 1 + (0.1)[1 - 0 + 2\cdot(1)] = 1 + 0.3 = 1.3$

$t_2 = 0.2$

$y_2 = y_1 + f(t_1, y_1) = 1.3 + (0.1)[1 - 0.1 + 2\cdot(1.3)] = 1.3 + 0.35 = 1.65$

$t_3 = 0.3$

$y_3 = y_2 + f(t_2, y_2) = 1.65 + (0.1)[1 - 0.2 + 2\cdot(1.65)] = 1.65 + 0.41 = 2.06$

Problem 2. (5 points) Consider the second-order differential equation:

$$2y'' - 3y' - 5y = 0.$$

Find the two solutions y_1 and y_2 for the equation by using the characteristic equation, compute the Wronskian $W(y_1, y_2)$, and then write down its general solution.

Solution: Characteristic equation

$$2r^2 - 3r - 5 = 0$$

$$(2r - 5)(r + 1) = 0$$

$$\Rightarrow r_1 = \frac{5}{2}, \quad r_2 = -1$$

$$\Rightarrow y_1(t) = e^{\frac{5}{2}t}, \quad y_2(t) = e^{-t}$$

$$W(y_1, y_2)(t) = \begin{vmatrix} e^{\frac{5}{2}t} & e^{-t} \\ \frac{5}{2}e^{\frac{5}{2}t} & -e^{-t} \end{vmatrix} = -e^{\frac{3}{2}t} - \frac{5}{2}e^{\frac{3}{2}t} = -\frac{7}{2}e^{\frac{3}{2}t} \neq 0$$

The general solution: $y(t) = c_1 e^{\frac{5}{2}t} + c_2 e^{-t}$